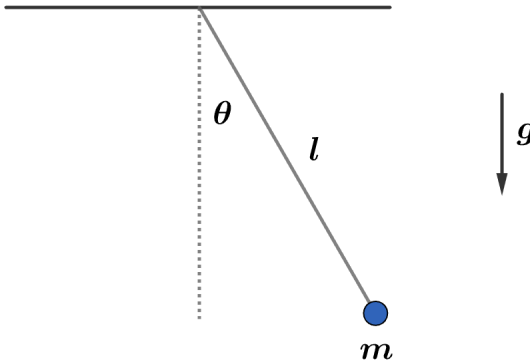


2011 F=ma Exam: Problem 10

Kevin S. Huang



Recall the period of a simple pendulum has the form (by dimensional analysis):

$$T = f(\theta) \sqrt{\frac{l}{g}}$$

- A) Decreasing l would decrease the period.
- B) Increasing m has no impact on the period.
- C) Increasing θ would increase T because the small angle approximation $\sin \theta \approx \theta$ breaks down. The actual restoring torque $\tau_{\text{act}} \propto \sin \theta$ is less than the assumed restoring torque $\tau_{\text{SHM}} \propto \theta$ as $\sin \theta < \theta$. This means the actual angular acceleration is less than assumed so it takes longer to return to the equilibrium.
- D) In an elevator accelerating upward, $g_{\text{eff}} > g$ so the period would decrease.
- E) In an elevator moving at constant speed, $g_{\text{eff}} = g$ so the period is unchanged.

Thus, the answer is C.